

No-Context Emoji Ablation Results

Validation Results

Model	Accuracy	Binary F1
Twitter (with emoji)	0.7514	0.4557
Twitter (no emoji)	0.6994	0.4222
Reddit (with emoji)	0.7300	0.7479
Reddit (no emoji)	0.7260	0.7439

Validation observations

For Twitter, removing emojis reduced F1 from **0.4557 to 0.4222**, suggesting emoji information may help sarcasm detection in validation.

For Reddit, the difference was minimal (**0.7479 vs 0.7439**), suggesting emojis contribute little.

Test Results

Model	Accuracy	Binary F1	Macro F1
Twitter (with emoji)	0.7341	0.4458	0.6354
Twitter (no emoji)	0.7399	0.5055	0.6645
Reddit (with emoji)	0.7160	0.7351	0.7145
Reddit (no emoji)	0.7175	0.7368	0.7160

Test observations

For Twitter, the no-emoji model performed better on the final test set, improving F1 from **0.4458 to 0.5055**.

For Reddit, performance remained nearly identical regardless of emoji preprocessing.

Main observations

1. **Reddit consistently outperformed Twitter by a large margin.**
One likely reason is that the Reddit dataset appears to be significantly larger, giving the model more training signal.
2. **Emoji impact appears dataset-dependent.**
Reddit shows almost no sensitivity to emoji removal.
3. **Twitter results are inconsistent between validation and test.**
Validation suggests emojis help, while test results suggest the opposite. This may suggest overfitting to emoji-specific patterns, or simply reflect instability/noise in the Twitter dataset.